Conserv Genet (2014) 15:757–770
DOI 10.1007/s10592-014-0576-1

RESEARCH ARTICLE

Range-wide comparisons of northern leatherside chub populations reveal historical and contemporary patterns of genetic variation

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Received: 3 June 2013 / Accepted: 22 January 2014 / Published online: 6 February 2014
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Abstract Patchily distributed species are those taxa whose populations occupy geographically insular habitats and their conservation often depends on an understanding of the relationship among disjunct populations. The objective of our study was to use molecular data and analytical techniques to separate the effects of historical and contemporary processes influencing the distribution of a high-desert minnow, the northern leatherside chub (*Lepidomeda copei*). Individuals from 23 populations were sequenced for 1,140 base pairs of the cytochrome B gene of the mitochondrial genome and genotyped at seven nuclear microsatellite loci. We estimated gene flow and examined population structure using both microsatellite and mtDNA data. Low sequence divergence and the distribution of shared haplotypes in multiple watersheds suggest historical connectivity between populations over a large geographic area. In contrast, patterns of microsatellite diversity indicate that populations of leatherside chub are isolated from one another with low levels of contemporary gene flow between populations. Our results suggest that populations of leatherside chub were historically more widely inter-connected and have recently been isolated, likely through a combination of natural and anthropogenic habitat fragmentation. As populations become increasingly isolated, they are more vulnerable to extirpation as a result of stochastic events. For northern leatherside chub, recent isolation and lack of gene flow among populations may affect their long-term survival in the arid landscapes of the Great Basin and surrounding watersheds because of widespread and increasing habitat alteration and fragmentation.

Keywords Leatherside chub · Population structure · Migration · Habitat fragmentation · Historical gene flow · Contemporary gene flow

Introduction

A combination of historical and contemporary forces affects the distribution and connectedness of populations across a landscape. Long-term climatic changes and natural vicariance represent historical mechanisms by which populations can become naturally isolated (Slatkin 1987; Morales et al. 2011). Such geographic isolation of populations is important as it can facilitate evolutionary processes, leading to differentiation among populations, changes in genetic diversity, and if isolation persists long enough, genetic divergence can lead to speciation (Slatkin 1987). In addition to long-term changes in the landscape that have isolated populations, the interaction of contemporary factors can also influence the distribution and isolation of populations. The mechanism of contemporary isolation can be highly variable, ranging from natural life history events that may effectively isolate populations during reproduction, to anthropogenic habitat alteration that can interfere with the movement of individuals among populations (Neville et al. 2006; Junker et al. 2012).

While natural and artificial barriers may isolate populations, it is often unclear how long insular populations have been isolated and whether such isolation has led to significant genetic differentiation. Similarly, a series of seemingly interconnected populations lacking any apparent geographic barriers may have little or no exchange of

Electronic supplementary material The online version of this article (doi:10.1007/s10592-014-0576-1) contains supplementary material, which is available to authorized users.

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individuals between populations. Molecular techniques have been invaluable in uncovering cryptic patterns of genetic differentiation in many organisms (Mock et al. 2006; Hughes et al. 2009). Mitochondrial DNA can uncover information about historical patterns of gene flow and colonization, providing a better understanding of population origins and divergence times among insular populations (Morales et al. 2011). Nuclear microsatellite DNA has been used to detect contemporary patterns of gene flow and genetic diversity (McCraney et al. 2010). By applying both measures, two major components of understanding intraspecific diversification can be captured and used as a basis for protecting the evolutionary history and the contemporary interconnectedness of a species.

Arid landscapes often contain rare and endemic species that occupy insular habitats distributed over large geographic areas (Cosacov et al. 2010; Guzik et al. 2012). Many of these flora and fauna are relicts of once widely distributed organisms and their current distribution has been shaped by events related to tectonism, volcanism, glaciation, and major changes in climatic and hydrologic conditions (Hershler and Sada 2002; Smith et al. 2002). The Great Basin region of western North America is among the most arid on the continent, but wetter periods during the late Pleistocene allowed for the widespread dispersal of flora and fauna (Hubbs and Miller 1948; Brown 1971; Dowling et al. 2002). As the landscape became more arid during the mid-Holocene, montane and aquatic habitats contracted, forcing species to seek refuge in the suitable habitats that remained (Meffe and Vrijenhoek 1988). Thus, once contiguous habitats are now small patches distributed across inhospitable desert scrub, where dispersal is infrequent (Fleishman et al. 2001; Floyd et al. 2005). Such isolation has sometimes led to significant population differentiation with little gene flow among neighboring habitats or even long term isolation leading to cryptic patterns of speciation (Mock et al. 2006; Guzik et al. 2012). Boreal mammals (Floyd et al. 2005), alpine plants (Billings 1978), butterflies (Fleishman et al. 2001), freshwater mollusks (Mock et al. 2004) and amphibians (Tanner 1978) are all examples of fauna that currently have localized populations in specialized habitats within the Great Basin region.

Over geologic time, changes in precipitation levels in arid regions have had a significant effect on the distribution and connectivity of populations inhabiting streams, lakes and springs. For aquatic fauna in arid environments, the alteration of connectivity between populations makes such species vulnerable to extirpation or extinction, as dispersal is already limited in these environments. In xeric landscapes, aquatic organisms may become adapted to extreme ecological conditions and are often the last remnants of well watered times (Meffe and Vrijenhoek 1988). Molecular investigations of fish fauna of the Great Basin suggest a complex history of isolation, punctuated by intermittent connectivity, with the origins of many lineages dating back to the Pliocene or Miocene (Dowling et al. 2002; Smith et al. 2002). The current distribution of species and the genetic relationships between populations were, in part, influenced by fluctuations, connections and desiccation of the large inland lakes that covered much of the western United States during the mid and late Pleistocene (Hubbs and Miller 1948; Johnson 2002; Billman et al. 2010; Houston et al. 2011). While historical processes may have created major subdivisions organizing biodiversity, over a much shorter time scale, contemporary habitat alterations can disrupt patterns of gene flow and metapopulation structure. Desert streams, in particular, have been negatively impacted by dams, irrigation diversions, culverts, and channelization, along with a reduction in habitat complexity from mining, grazing, urbanization and agriculture (Meffe 1986; Meffe and Vrijenhoek 1988). Separating the historical patterns of diversification from the more recent effects of anthropogenic habitat alteration is critical to understanding the evolutionary relationships between populations and to preserving the genetic diversity and connections that remain.

Populations of northern leatherside chub (*Lepidomeda copei*) present an opportunity to investigate historical and contemporary genetic patterns of a rare and patchily distributed high-desert fish species. The range of northern leatherside chub can be generally characterized as high-desert or cold-desert where the vast majority of moisture comes in the form of winter precipitation and plant communities are dominated by sagebrush steppe habitat in low and mid elevations (Comstock and Ehleringer 1992). Extant populations of this minnow occupy small 'islands' of habitat within three major watersheds: the Bear River watershed of the northern Bonneville basin, the Snake River watershed of the Columbia River basin and the Green River watershed of the Colorado River basin (Wallace 1980; Zafft et al. 2009).

In our study, we compare northern leatherside chub populations from across their geographic range to examine the degree of isolation and potential gene flow in the Great Basin and adjacent exorheic basins. To distinguish historical connections and colonization patterns from contemporary gene flow between populations, we analyzed patterns of genetic diversity using mitochondrial and nuclear DNA markers from populations throughout the range of northern leatherside chub.

Materials and methods

Study sites

To describe historical and contemporary genetic diversity in northern leatherside chub (hereafter referred to as

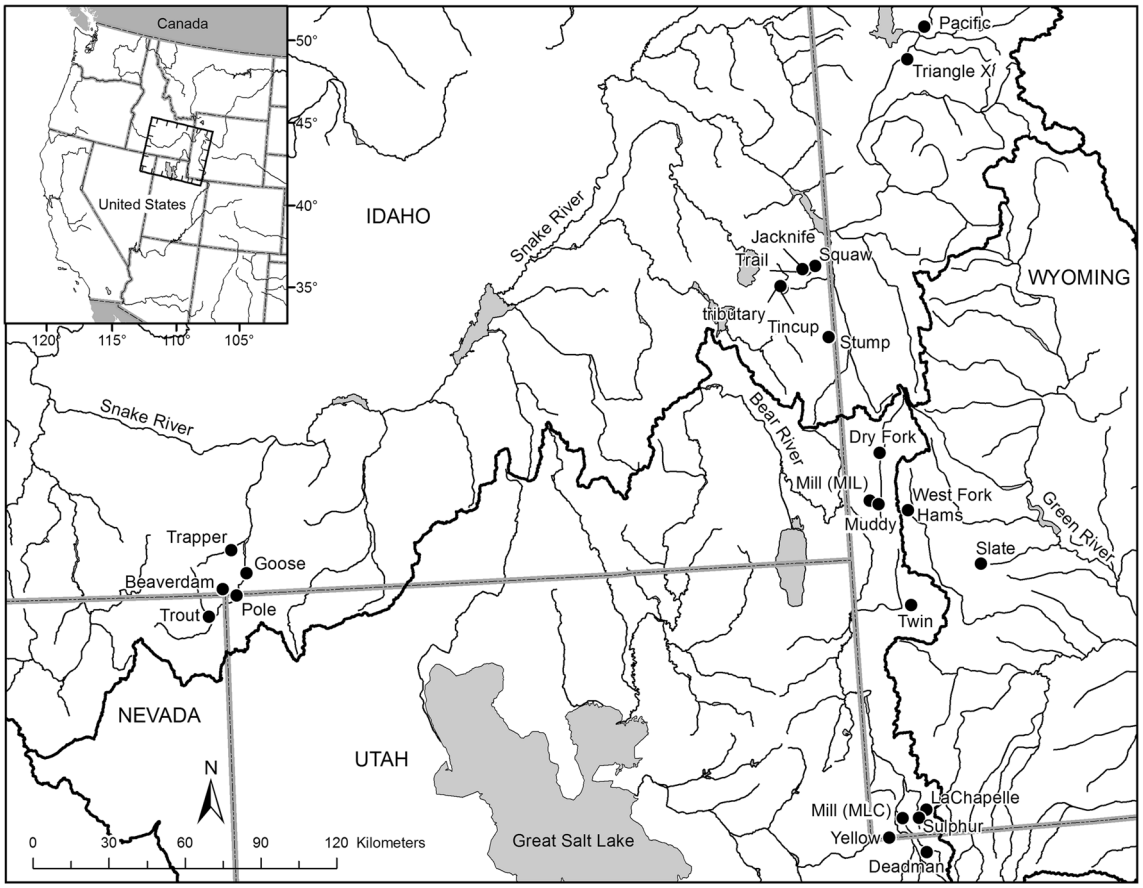

Fig. 1 Distribution of northern leatherside chub populations collected for this study. *Bold lines* denote watershed boundaries between the Snake River, Green River, and the Bonneville basin (the Bear River lies within this endorheic watershed). *Inset map* indicates the location of the study area in the western United States

leatherside chub), we attempted to collect samples from all known populations based on historical records from published papers, technical reports and museum specimens. The geographic distribution of the species encompasses portions of the Snake River, Bear River, and Green River watersheds within Idaho, Nevada, Utah and Wyoming (Fig. 1, Supplemental Fig. 1).

Tissue samples from leatherside chub were obtained from 2009 to 2011 using two methods: backpack electrofishing and baited minnows traps. We used two electrofishing strategies to sample streams for leatherside chub. The first method was to sample random 100 m stream reaches within stream segments where leatherside chub have been observed; surveying all available habitat within a reach. The second method consisted of selectively sampling stream reaches up to 3 km in length, where only the most appropriate habitats were sampled based on our observations of the species affinity for habitat complexity. Minnow traps were used when backpack electrofishing was ineffective (e.g., deep pools, beaver ponds). For each leatherside chub captured, tissue was taken from a small portion of the pelvic fin and was stored in 95 % ethanol. Fish were then released back into the stream near the point of capture. In the laboratory, genomic DNA was extracted using a Chelex extraction protocol (Blakney 2012).

Mitochondrial DNA sequencing and analyses

For mtDNA analyses, we sequenced a 1,140 base pair (bp) segment of the mitochondrial cytochrome B gene (cytB) using external primers LA and HA (Johnson and Jordan 2000; Dowling et al. 2002). The PCR amplification was carried out in a 25 µl reaction containing 10.5 µl·dH_2O, 10 µl Reddy Mix (Thermo Scientific Inc., Waltham, MA), 1.25 µl forward and reverse primers, 1 µl·$MgCl_2$, and 1 µl·DNA. The thermal profile consisted of denaturation at 94 °C for 1 min, annealing at 48 °C for 1 min, and extension at 72 °C for 2 min for 35 cycles and a final extension at 72 °C for 30 min. Forward and reverse sequences were generated using an ABI 3738xl DNA analyzer at the University of Washington High-Throughput Genomics Unit.

Leatherside chub mtDNA sequences were edited using Sequencher software (v4.9; Gene Codes Corporation, Ann

Arbor, MI) and aligned in MUSCLE (Edgar 2004). Additional *Lepidomeda* sequences were downloaded from GenBank (www.ncbi.nlm.nih.gov/genbank/) and also included in the alignment. The number of different mtDNA haplotypes as well as estimates of haplotype and nucleotide diversity were calculated using DnaSP (v5; Rozas 2009). We constructed a minimum spanning network using mtDNA haplotypes in Arlequin (v3.5; Excoffier and Lischer 2010). Patterns of genetic diversification were estimated by calculating average pairwise sequence divergence among individuals, populations and watersheds using MEGA (v5.05; Tamura et al. 2011). To examine sequence divergence at different spatial scales, individuals were grouped into populations based on sampling locations and then populations were grouped into subbasins within watersheds (Table 1). Mitochondrial DNA diversity was examined using an Analysis of Molecular Variance (AMOVA) in Arlequin, partitioning pairwise differences between sequences among subbasins, among populations within subbasins, and within populations.

Microsatellite amplification and analyses

We identified seven polymorphic microsatellite loci using primers developed for leatherside chub: 28-7, 4-6, 25-8, 27-8 (Mock et al. 2008), as well as other cyprinid fishes: OCR 113, OCR 112 (Ardren et al. 2007) and GEL 225 (Keeler-Foster et al. 2004). Microsatellite amplification was conducted in a 15 µl reaction consisting of 6 µl Reddy Mix master mix (Thermo Scientific Inc., Waltham, MA), 5 µl·dH_2O, 2 µl·DNA, and 1 µl of each fluorescently labeled forward and reverse primer. PCR conditions consisted of an initial denaturation at 95 °C for 3 min followed by 35 cycles at 94 °C for 30 s, 48–54 °C for 30 s, 72 °C for 1 min, and a final extension at 72 °C for 30 min. PCR products were submitted to the Idaho State University Molecular Research Core Facility and genotyped using an ABI 3130 sequencer and GeneMapper software (v4.0; Applied Biosystems, Foster City, CA).

Microsatellite diversity was measured by the number of alleles per locus (A), observed heterozygosity (H_O), expected heterozygosity (H_E), allelic richness (A_r) using rarefaction, and inbreeding coefficient (F_{IS}) in Arlequin and FSTAT (v2.9.3; Goudet 2001). Tests for significant differences in allelic richness, observed heterozygosity, and expected heterozygosity between populations were conducted using 10,000 permutations in FSTAT. We tested microsatellite genotypes for deviations from Hardy–Weinberg equilibrium at each locus within each population. Critical significance levels were evaluated using a Bonferroni correction to adjust for multiple comparisons. Linkage disequilibrium was assessed across all pairs of loci for all populations using 9,960 permutations. Analysis for deviations from Hardy–Weinberg equilibrium and linkage disequilibrium were both conducted using the software program FSTAT. To examine the distribution of microsatellite variation, an AMOVA, as implemented in Arlequin, was used to partition allelic frequencies among subbasins, among populations within subbasins, and within populations.

We also examined fine-scale genetic structuring among populations within and between subbasins. Pairwise genetic differentiation (F_{ST}) estimates were calculated in Arlequin with significance based on 10,000 permutations. The relationship between geographic distance and genetic differentiation was examined by testing for a pattern of isolation by distance using the IBD web service based on 10,000 randomizations (Bohonak 2002). Geographic distances between populations within watersheds were estimated by measuring stream distance between populations with MapSource software (v6.12.4; Garmin Ltd., Olathe KS, USA).

Population genetic structure of leatherside chub was also analyzed using a principal components analysis (PCA) and a Bayesian assignment of individuals to populations. The placement of populations on the principal components axis is based on the genetic similarities across loci and was calculated in SAS software (v9.2). Structure (v2.2; Pritchard et al. 2000) was used to assess the most likely number of populations (K) using an individual-based Bayesian assignment method. Five independent runs for each K (3–25) were conducted using an admixture model, run for 500,000 iterations with a burn-in of 200,000 iterations. Our Structure analyses were performed using the admixture and correlated allele frequencies models and without prior population information. The most likely number of population clusters (K) was determined using the log-likelihood of the posterior probability [L(K)] and the estimation of ΔK (Evanno et al. 2005).

To compare contemporary and historical rates of migration, we used the software programs BayesAss (v3.0.1; Wilson and Rannala 2003) and Migrate (v3.3.2; Beerli 2006) respectively. BayesAss uses a Bayesian Markov-chain Monte Carlo (MCMC) approach and a genetic assignment method to estimate contemporary rates of migration within the last five generations (Wilson and Rannala 2003). In contrast, Migrate uses the coalescent and MCMC techniques to estimate long-term effective population sizes and gene flow (Beerli 2006) over much larger time scales (1000s of years, Chiucchi and Gibbs 2010). The BayesAss analyses included three independent runs of 10 million iterations with a 1 million generation burn-in, sampling the chain every 1,000 generations to estimate migration in the last one to two generations (~4–10 years for leatherside chub). For Migrate analyses, we used a Bayesian approach to estimate migration between

Table 1 Location of core collection areas for 24 streams inhabited by northern leatherside chub as well as sample size for microsatellite [n(msat)] and mitochondrial analyses [n(mtDNA)], observed heterozygosity (H_O), allelic richness (A_R), number of alleles, inbreeding coefficient (F_{IS}), number of haplotypes (H), haplotype diversity (h) and nucleotide diversity (π). See Fig. 1; Supplemental Table 1 for additional location information

Population, State	Watershed	Subbasin	n (msat)	H_O (mean)	A_R (mean)	No. of alleles (mean)	F_{IS}	n (mtDNA)	H	h	π
Dry Fork Smiths Fork, WY	Bear River	Middle Bear	61	0.72	3.45	6.86	0.03	30	1	0.00	0.0000
Mill Creek (MIL), WY	Bear River	Middle Bear	15	0.75	4.10	8.43	0.10	1	1	0.00	0.0000
Muddy Creek, WY	Bear River	Middle Bear	36	0.67	4.26	11.29	0.20	4	3	1.00	0.0020
Twin Creek, WY	Bear River	Middle Bear	33	0.68	3.92	11.14	0.13	4	2	0.50	0.0005
Deadman Creek, UT	Bear River	Upper Bear	60	0.72	3.70	8.57	0.08	10	2	0.53	0.0005
LaChapelle Creek, WY	Bear River	Upper Bear	35	0.74	4.03	11.29	0.07	3	1	0.00	0.0000
Mill Creek (MLC), WY	Bear River	Upper Bear	29	0.72	4.37	12.0	0.15	4	4	1.00	0.0025
Sulphur Creek, WY	Bear River	Upper Bear	37	0.78	4.42	12.86	0.09	8*	3	0.71	0.0019
Yellow Creek, UT	Bear River	Upper Bear	57	0.73	4.33	14.14	0.12	8	2	0.25	0.0003
Slate Creek, WY	Green River	Green River	40	0.60	3.00	5.43	0.07	8	3	0.52	0.0006
West Fork Hams Fork, WY	Green River	Green River	39	0.55	2.83	5.71	0.08	8	1	0.00	0.0000
Beaverdam Creek, ID	Snake River	Middle Snake	69	0.66	3.74	10.71	0.10	22*	4	0.33	0.0007
Goose Creek, ID	Snake River	Middle Snake	7	0.67	4.04	6.43	0.15	7	3	0.76	0.0011
Pole Creek, ID/UT	Snake River	Middle Snake	24	0.70	3.94	9.86	0.08	14	3	0.62	0.0010
Trapper Creek, ID	Snake River	Middle Snake	64	0.71	4.15	14.0	0.10	12	3	0.68	0.0008
Trout Creek, NV	Snake River	Middle Snake	29	0.61	4.22	12.71	0.23	10	2	0.47	0.0005
Pacific Creek, WY	Snake River	Upper Snake	39	0.65	3.82	9.14	0.18	11*	4	0.71	0.0009
Jackknife Creek, ID	Snake River	Upper Snake	65	0.65	3.56	10.0	0.10	12	3	0.56	0.0011
Squaw Creek, ID	Snake River	Upper Snake	60	0.62	3.60	10.71	0.13	12	3	0.32	0.0008
Stump Creek ID	Snake River	Upper Snake	7	0.58	3.29	4.43	0.20	7	2	0.29	0.0003
Tincup Creek, ID	Snake River	Upper Snake	19	0.54	3.48	7.14	0.23	6	2	0.33	0.0007
Trail Creek, ID	Snake River	Upper Snake	33	0.68	3.67	8.29	0.09	10	2	0.36	0.0007
unnamed tributary, ID	Snake River	Upper Snake	23	0.61	3.46	7.43	0.15	6	3	0.73	0.0014
Triangle X Creek, WY	Snake River	Upper Snake	2	–	–	–	–	2	1	0.00	0.0000

* Includes samples from Genbank database, see text for accession numbers

watersheds. We conducted three independent runs using a Brownian mutation model: two runs at 1 million generations and a final run with 10 million generations. To compare estimates of historical and contemporary gene flow, mutation adjusted migration values ($M = m/\mu$) from Migrate were used to estimate migration rates (m) by multiplying all M values by the mutation rate (μ), where $\mu = 5.4 \times 10^{-4}$ (Apodaca et al. 2012).

Results

Leatherside chub were collected from populations within the Snake River, Bear River, and Green River watersheds (Supplemental Table 1; Fig. 1). For the Snake River watershed, populations were sampled from five streams in the Goose Creek area of southern Idaho and northern Nevada-Utah, from six streams in the Salt River area of eastern Idaho, and from Pacific Creek in western Wyoming (Fig. 1). Two specimens were also collected from Triangle X Creek, a small tributary that drains directly into the main-stem Snake River, but were excluded from microsatellite analyses due to low sample size. Populations in the upper Bear River subbasin were sampled from five streams in northeastern Utah–southwestern Wyoming. Populations were also sampled from four streams in the middle Bear River subbasin in western Wyoming. In the Green River watershed, populations are known from only two tributaries both of which are upstream of Flaming Gorge Reservoir in the Green River watershed of western Wyoming. For analytical comparisons we considered each of the above areas as subbasins of the Snake River, Bear River or Green River watersheds (Table 1; Fig. 1).

Mitochondrial DNA analyses

A total of 199 leatherside chub were sequenced for 1,140 bp of the cytB gene, representing individuals from each population sampled (Table 1; GenBank accession numbers KJ175008–KJ175027). An additional 26 cytB sequences were downloaded from GenBank that had location data available and were also included in our analyses (AF270885–AF270893, Johnson and Jordan 2000; AF452086–AF452087, Dowling et al. 2002; AY825431–AY825445, Johnson et al. 2004). Of the 225 leatherside chub sequences compiled, we identified 24 unique haplotypes.

Across all sampling locations, mtDNA sequence data indicated close relationships between populations. Overall, average pairwise sequence divergence across all populations was low (0.35 %). Average haplotype divergence among subbasins ranged from 0.21 % between the upper and middle Bear River to 0.52 % between the upper Snake and middle Snake River (Supplemental Table 2). Based on an AMOVA, differences among subbasins accounted for 43.7 % of the variation in the cytB sequence diversity (Table 2a). Variation among populations within subbasins accounted for 20.3 % of the diversity, while within population variation explained 35.9 % of the sequence diversity (Table 2a).

The minimum-spanning network of the 24 unique haplotypes revealed limited geographic structure among leatherside chub populations according to their distribution in the network (Fig. 2). Six haplotypes were observed only in Salt River populations, but did not form a unique cluster in the network. Similarly, three of four haplotypes observed in Pacific Creek, a tributary to the upper Snake River, were only found in that location and did not form a distinct cluster (Fig. 2). The fourth haplotype (Hap 13) was not only found in Pacific Creek, but also in the Bear River and the Green River populations and was the most common haplotype across those areas (Fig. 2). The two fish sampled from Triangle X Creek also shared the most common haplotype detected in Pacific Creek (Hap 13). Nine haplotypes were observed in the Bear River, six of which were only found in that watershed (Fig. 2). Three haplotypes were observed in the Green River populations, all of which were also found in the Bear River populations, including two rare haplotypes (Hap 14 and 15). The one exception to the limited geographic structuring of mtDNA haplotypes was the six haplotypes observed in the Goose Creek populations. Based on the number of mutational steps, these haplotypes were closely related to haplotypes from the Bear River, but were found only in the Goose Creek populations and grouped together in the haplotype network (Fig. 2).

Table 2 Analysis of molecular variance (AMOVA) between groups of populations of northern leatherside chub from five subbasins of the Bear River, Snake River, and Green River watersheds. Genetic variance was estimated between watersheds, within watersheds, and within populations based on (a) mtDNA pairwise sequence diversity and (b) nuclear microsatellite allele frequencies

Source of variation	df	Sum of squares	Percent variation	*P* value
(a)				
Among subbasins	4	378.18	43.72	<0.001
Among populations within subbasins	19	169.91	20.34	<0.001
Within populations	195	308.80	35.93	<0.001
Total	218	850.89		
(b)				
Among subbasins	4	312.69	5.67	<0.001
Among populations within subbasins	18	281.11	5.85	<0.001
Within populations	1,765	4,704.09	88.49	<0.001
Total	1,787	5,297.89		

Microsatellite analyses

We scored seven microsatellite loci for 894 leatherside chub collected from 23 streams representing the three major watersheds where this species is found. All microsatellite loci were variable, with the number of alleles per locus ranging from five (25-8) to 22 (OCR 113) (Table 1). Tests revealed no significant deviation from Hardy–Weinberg equilibrium or linkage disequilibrium, thus all loci and populations sampled were used in the analyses. Population sample size ranged from seven to 69 individuals, with an average of 38 individuals per population (Table 1). Mean observed heterozygosity ranged from 0.54 in Tincup Creek to 0.78 in Sulphur Creek, with a mean of 0.67 for all populations. Allelic richness ranged from 2.83 in West Fork Hams Creek to 4.42 in Sulphur Creek, with a mean of 3.80 for all populations. The average number of alleles per locus ranged from 4.43 in Stump Creek to 14.14 in Yellow Creek, with an overall average of 9.50 alleles per locus (Table 1).

At the watershed level, the Bear River populations displayed significantly higher levels of average observed heterozygosity ($H_o = 0.72$) relative to populations from the Snake River ($H_o = 0.65$) and Green River ($H_o = 0.57$; $P = 0.006$). Allelic richness was also significantly higher in Bear River leatherside chub (Ar = 4.06) when compared with fish from the Snake River (Ar = 3.75) or Green River (Ar = 2.92; $P < 0.001$). Observed heterozygosity ($P = 0.006$) and allelic richness ($P < 0.001$) were also significantly higher in leatherside chub from the Snake River relative to the Green River populations. Across the

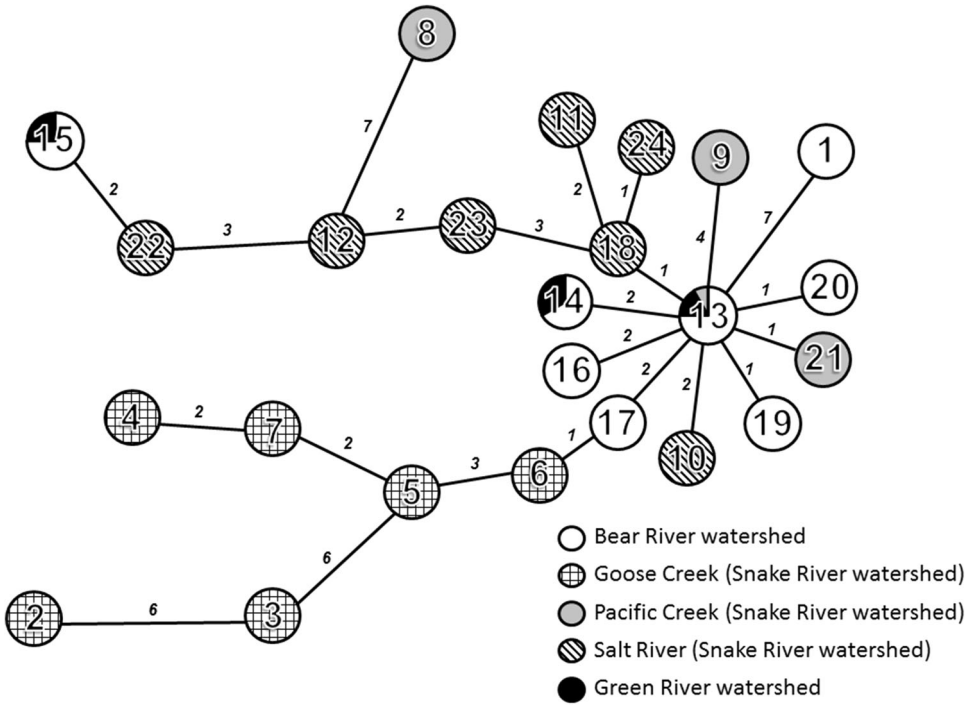

Fig. 2 Minimum spanning network of 24 cytochrome B haplotypes in northern leatherside chub based on their distribution in five different watersheds. *Numbers in the circles* represent haplotype numbers. *Numbers on lines* indicate the number of mutational changes

species geographic range, an analysis of molecular variance revealed 5.7 % of genetic variation occurs among subbasins, 5.9 % of genetic variation occurs among population within subbasins, while 88.5 % of genetic variation exists within individual populations (Table 2b).

Overall, F_{ST} estimates indicate significant genetic differentiation between almost all populations within watersheds. Most populations in the Snake River watershed were significantly different from one another (F_{ST} = 0.026–0.166) except for a few populations in Salt River streams from the upper Snake River subbasin (Supplemental Table 3a; Fig. 1). F_{ST} values indicated that Jackknife Creek was not significantly different from Stump Creek or Trail Creek (Jackknife Creek tributary). However, Jackknife Creek was significantly different from its other tributary known to contain leatherside chub, Squaw Creek (F_{ST} = 0.026). The unnamed tributary, the only Tincup Creek tributary where leatherside chub have been found, was not genetically different from Tincup Creek proper or from Stump Creek. All Goose Creek populations from the middle Snake River subbasin were significantly different from each other. Most Bear River populations were significantly different from one another (F_{ST} 0.012–0.125), except Mill (MLC) and Sulphur Creeks from the upper Bear River subbasin (Supplemental Table 3b; Fig. 1). The two Green River populations, West Fork Hams Fork and Slate Creek, were significantly different from one another (F_{ST} = 0.154).

There was a significant pattern of isolation by distance in both the Snake River and Bear River populations (Fig. 3); however, the correlation between genetic distance and geographic distance was stronger in the Snake River (Mantel r = 0.74, $P < 0.001$) when compared to the Bear River (Mantel r = 0.35, $P = 0.006$). In addition to comparisons at the watershed level, we also examined isolation by distance at the subbasin level. We detected a significant pattern of isolation by distance in the upper Snake River subbasin (Mantel r = 0.77, $P = 0.007$). In contrast, in the middle Snake subbasin, we found no relationship between geographic distance and genetic distance (Mantel r = −0.59, $P = 0.88$). Similarly, no significant relationship was detected in the upper Bear River subbasin (Mantel r = 0.113, $P = 0.29$) or the middle Bear River subbasin (r = 0.92, $P = 0.13$).

Principal components analysis and Bayesian clustering methods revealed similar patterns of population structuring. The first three axes of the PCA explained 46.7 % of the genetic variation, PC axis 1 19.6 %, PC axis 2 15.0 %, and PC axis 3 11.1 % (Fig. 4). The first axis separated the Snake River from the Bear River and Green River watersheds, while the second axis separated populations between the upper and middle Snake River (Fig. 4a). The third axis separated Green River populations from the Bear River populations and separated Pacific Creek from the other Snake River populations (Fig. 4b). When we treated each stream where fish were collected as an independent population, the likelihood of the posterior probability [L(K)] reached an asymptote at K = 7 (Fig. 5). Likewise, the ΔK statistic indicated that the data could be organized into seven population clusters (Fig. 5). Three population

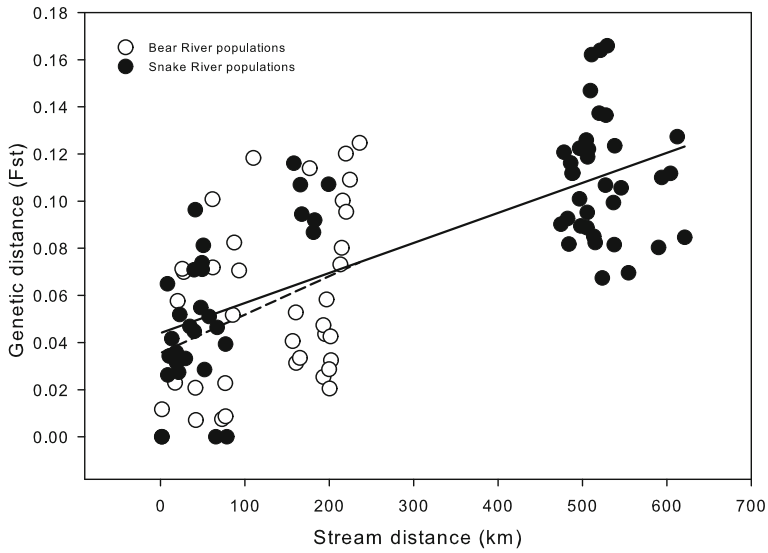

Fig. 3 Relationship between stream distance (km) and genetic distance (F_{ST}) for northern leatherside chub populations in the Snake River watershed (*closed circles* Mantel r = 0.744, $P < 0.001$) and Bear River watershed (*open circles*, Mantel r = 0.348, $P = 0.006$)

clusters described three areas of the Snake River where we collected samples: Goose Creek streams, Salt River streams, and Pacific Creek. The two Green River tributaries also grouped into a single cluster. The remaining three clusters grouped Bear River populations. The largest Bear River cluster contained six populations: LaChapelle Creek, Sulphur Creek, Yellow Creek, Twin Creek, Mill Creek (MLC) and Muddy Creek. The second cluster contained two populations Dry Fork and Mill Creek (MIL), while the third cluster contained only the Deadman Creek population.

A comparison of contemporary and historical migration rates revealed very different movement patterns between the two periods. Bayesian estimates of contemporary gene flow (m_c) indicate very low levels of recent migration between most pairwise comparisons of subbasins (Table 3). Overall, contemporary estimates of migration averaged only 0.00913 and 19 of 20 lower bounds of the 95 % confidence intervals overlapped zero (Table 3). The one comparison that revealed much higher levels of recent migration (0.0875) occurred between the upper Bear River and middle Bear River subbasins. In contrast, our adjusted estimates of historical gene flow (m_h) indicate much higher levels for leatherside chub within and between subbasins. Average estimates of historical migration were 0.0711 with no lower 95 % confidence intervals overlapping zero (Table 3). When paired by subbasin, historical levels were 80.73 % higher than contemporary estimates of migration (paired t = 9.43, $P < 0.0001$).

Discussion

In this study we conducted a range-wide comparison of northern leatherside chub to determine how historical and contemporary factors may have shaped the genetic population structure of this species. Extant populations of leatherside chub are patchily distributed and largely isolated from one another across their range in the Great Basin and adjacent exorheic watersheds. Results of our study suggest that this pattern is a reflection of historical and contemporary processes. The distribution of mtDNA haplotypes, in conjunction with geologic evidence, infers historical hydrologic connectivity between populations in the Snake River, Bear River, and Green River watersheds. Evidence of connections between these watersheds was inferred by the presence of a single dominant haplotype shared between populations in the Bear and Green Rivers, and in Pacific Creek of the upper Snake River. In contrast, microsatellite analyses coupled with records from fish distribution surveys indicate that populations are currently isolated from one another. Taken together, the mtDNA and microsatellite results reveal a pattern of historical connections and recent isolation and genetic differentiation.

Historical connections: genetics and geology

Based on mtDNA data for leatherside chub from this study, low sequence divergence and the presence of haplotypes across multiple watersheds, our data indicate that three geographically isolated watersheds where leatherside chub are found were once connected. These watersheds include: the Bear River of the northern Bonneville basin, the Snake River of the Columbia River basin and the Green River of the Colorado River basin. As an ancient tributary to the Snake River, the Bear River is thought to share part of its history with the Snake River until about 600,000–50,000 years ago when lava flows diverted the Bear River southward towards the Bonneville

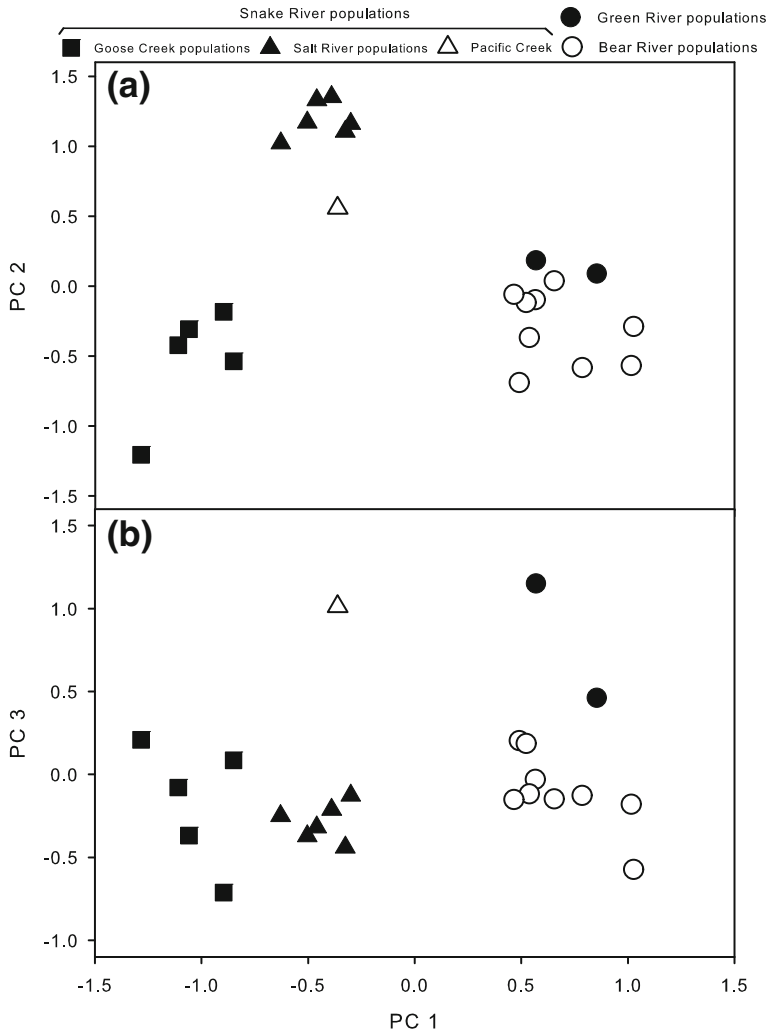

Fig. 4 Principal component scores for northern leatherside chub populations based on microsatellite variation at seven polymorphic loci for (**a**) principal component (PC) axis 1 versus PC axis 2 or (**b**) PC axis 1 versus PC axis 3. Populations from various areas are denoted by the following *symbols*: Salt River populations (*closed triangles*), Pacific Creek (*open triangle*), Goose Creek populations (*closed squares*), Green River populations (*closed circles*), and Bear River populations (*open circles*)

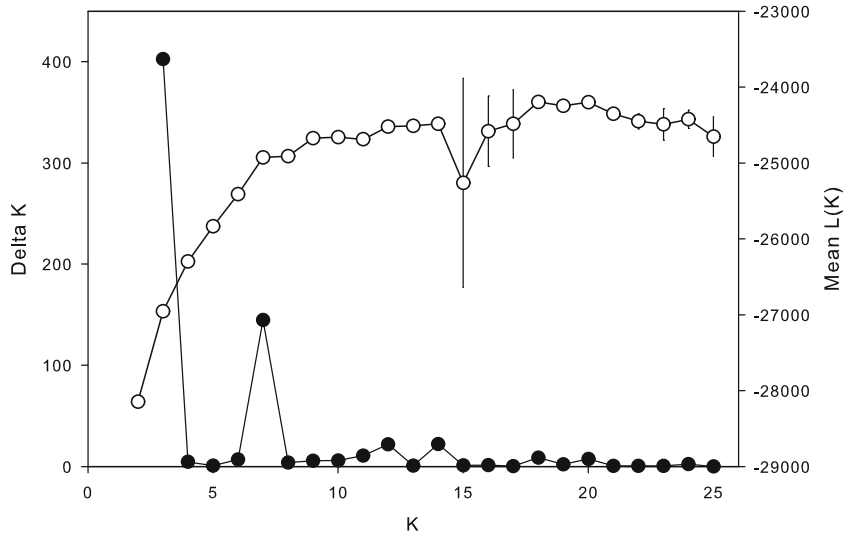

Fig. 5 The average log likelihood ± 1 SD (L(K), *open circles*) and delta K (*closed circles*) estimates for 3–25 clusters (K) of northern leatherside chub

basin (Bouchard et al. 1998; Link et al. 1999). During this early period of diversion the Bear River was intermittently connected to the Bonneville basin (Link et al. 1999), allowing the transfer of fauna between the two watersheds (Hubbs and Miller 1948; Smith 1978). The connections of these two watersheds is reflected in patterns revealed by

Table 3 Mean pairwise estimates of contemporary (m_c) and historical (m_h) migration rates and 95 % confidence intervals for northern leatherside chub populations grouped according to five subbasins in the Snake River, Bear River, and Green River watersheds

Subbasin comparison	95 % CI			95 % CI		
	m_c	Lower	Upper	m_h	Lower	Upper
Middle Snake–upper Snake	0.0041	−0.0014	0.0083	0.0433	0.0299	0.0562
Middle Snake–middle Bear	0.0038	−0.0027	0.0098	0.1091	0.0940	0.1292
Middle Snake–upper Bear	0.0095	−0.0017	0.0169	0.0571	0.0461	0.0677
Middle Snake–Green	0.0032	−0.0025	0.0086	0.0918	0.0778	0.1055
Upper Snake–middle Snake	0.0047	−0.0008	0.0083	0.1049	0.0929	0.1166
Upper Snake–middle Bear	0.0019	−0.0016	0.0053	0.1016	0.0871	0.1159
Upper Snake–upper Bear	0.0033	−0.0012	0.0068	0.0953	0.0832	0.1069
Upper Snake–Green	0.0032	−0.0011	0.0065	0.1312	0.0983	0.1584
Middle Bear–middle Snake	0.0027	−0.0026	0.0080	0.0462	0.0346	0.0572
Middle Bear–upper Snake	0.0041	−0.0026	0.0101	0.0304	0.0194	0.0407
Middle Bear–upper Bear	0.0875	0.0510	0.0551	0.0511	0.0396	0.0619
Middle Bear–Green	0.0176	−0.0040	0.0326	0.1016	0.0806	0.1231
Upper Bear–middle Snake	0.0022	−0.0019	0.0062	0.0915	0.0742	0.1091
Upper Bear–upper Snake	0.0023	−0.0020	0.0065	0.0550	0.0385	0.0727
Upper Bear–middle Bear	0.0053	−0.0029	0.0124	0.0753	0.0598	0.0839
Upper Bear–Green	0.0032	−0.0025	0.0086	0.1299	0.1076	0.1595
Green–middle Snake	0.0039	−0.0037	0.0115	0.0292	0.0176	0.0403
Green–upper Snake	0.0079	−0.0031	0.0166	0.0131	0.0025	0.0234
Green–middle Bear	0.0041	−0.0037	0.0118	0.0379	0.0245	0.0515
Green–upper Bear	0.0081	−0.0029	0.0166	0.0256	0.0126	0.0385

molecular investigations of bluehead sucker (*Catostomus discobolus*; Hopken et al. 2013), Utah sucker (*Catostomus ardens*; Mock et al. 2006), cutthroat trout (*Oncorhynchus clarkii* ssp; Loxterman and Keeley 2012), speckled dace (*Rhinichthys osculus*; Billman et al. 2010), Utah chub (*Gila atraria*; Johnson 2002) and mountain whitefish (*Prosopium williamsoni*; Whiteley et al. 2006). In each of these cases, populations from the upper Snake and Bear rivers formed a single genetic cluster distinct from all other populations. Likewise, the most common mtDNA haplotype in Pacific Creek, an upper Snake River tributary, was also found in Bear River leatherside chub and was the only haplotype shared between the two basins. Using an average estimated rate of sequence divergence of 1.17 % per million years for the cytB gene in cyprinid fishes from the Snake River and Bonneville watersheds (Dowling et al. 2002; Smith et al. 2002), average haplotype divergence between these two watersheds for leatherside chub indicates a separation between 258,000 and 407,000 years ago. This estimate coincides with the timing of the diversion of the Bear River from the Snake River and lends more evidence of faunal transfer between the two areas that now drain separate watersheds.

In addition to the evidence for a connection between the Bear River and the Snake River, the mtDNA results also suggest a connection between the Bear River and Green River watersheds. The haplotype that was dominant in populations from both Pacific Creek and the Bear River was also dominant in the two Green River populations. While there has been anecdotal evidence that these populations are introduced (Zafft et al. 2009), the occurrence of several other fish species native to the Snake River, Bear River, and Green River watersheds, including bluehead sucker, mountain sucker (*Catostomus platyrhynchus*), speckled dace, mountain whitefish and sculpin (*Cottus* spp.), strongly suggests a historical connection between these watersheds (Hansen 1985; Dowling et al. 2002). Low passes along drainage divides, orientation of present day tributaries, glacial moraines, along with the presence of gravel terraces and thrust faults indicate that there were numerous hydrologic connections between the Snake River and Green River, as well as the Bear River and Green River, by way of headwater transfer from stream captures and receding glaciers during the Pleistocene and early Holocene (Rubey et al. 1975; Hansen 1985). Such headwater transfers are thought to be an important mechanism explaining the distribution of fish species across large

geographic areas (Smith 1978; Minckley et al. 1986; Hughes et al. 2009). Unlike other studies that have detected at least some genetic signature of a recent anthropogenic introduction for a fish species (Kinziger et al. 2011), measures of heterozygosity, allelic richness, inbreeding, haplotype diversity, and nucleotide diversity of Green River leatherside chub were similar to measures for populations in the Snake and Bear rivers. When combined with the occurrence of other native fish fauna in both watersheds, it is possible that northern leatherside chub may have naturally entered the Green River watershed with other Snake-Bonneville fauna. Given the limited distribution of this species, Green River populations should be protected for conservation purposes at a minimum.

Contemporary isolation: genetics and fragmentation

Although leatherside chub were distributed in portions of three major watersheds in North America by historical changes in hydrology, geology and climate, their current distribution within each of these areas has been restricted to a few small, isolated streams. Interestingly, while heterozygosity estimates for populations of leatherside chub were similar to those reported in other populations of minnows (Kinziger et al. 2011; DeHaan et al. 2012; O'Malley et al. 2013), allelic richness was much lower in leatherside chub. In fact, allelic diversity in leatherside chub (Table 1) was lower than other minnow species with restricted distributions (Ar = 5.5, *Notropis mekistocholas*, Saillant et al. 2004; Ar = 9.8, *Oregonichthys crameri*, DeHaan et al. 2012), as well as more widely distributed species ($A_r = 5.9$, *Rhinichthys osculus*, Kinziger et al. 2011). Results from our microsatellite analyses indicate significant population differentiation, even between streams within the same area. Several lines of evidence suggest that such isolation is probably related to contemporary habitat fragmentation that now separates populations. In contrast to the low sequence divergence and wide distribution of mtDNA haplotypes across watersheds, the frequency of mtDNA haplotypes and microsatellite alleles indicate significant genetic differentiation and organization at the subbasin level. The AMOVA, PCA, and STRUCTURE analyses organized leatherside chub populations by subbasins across all three watersheds. For both the Snake River and Bear River watersheds, F_{ST} values further revealed significant genetic differentiation within subbasins for almost all populations sampled. High levels of genetic differentiation between neighboring populations is not surprising as many extant populations are separated by widespread habitat fragmentation. With the agricultural development of the American West beginning approximately 150 years ago, many parts of this region have experienced habitat alteration from irrigation diversions of stream flow, impoundment from dams, as well as impacts on water quality from land-use practices in adjacent terrestrial habitats (Denton 2007). Similarly, during the latter half of the 20th Century, the widespread use of itchytoxins to remove non-game fishes in many Snake River tributaries may have completely eliminated leatherside chub from some areas (Horton 1992); while the practice of introducing numerous non-native predators has probably negatively impacted leatherside chub in other areas (Walser et al. 1999; Nannini and Belk 2006).

Both the pattern of isolation by distance and the AMOVA indicate that populations of leatherside chub are geographically structured, with stream distance explaining a significant proportion of the variation in genetic distance between populations. However, the presence of gaps at intermediate distances in our analysis of isolation by distance suggests that populations at intermediate distances were at one time present and have been recently extirpated, or were not sampled (Keeley et al. 2012). Previous distribution surveys of fish species indicate that leatherside chub were more widespread than we were able to detect (Hubbs and Miller 1948; Zafft et al. 2009). For example, leatherside chub were recorded more widely in the South Fork of the Snake River (Hopkirk and Behnke 1966), the Little Wood River (Hubbs and Miller 1948) as well as in Ross Fork Creek (Johnson et al. 2004) and Buffalo Fork (Zafft et al. 2009), all of which are locations of intermediate distance in the Snake River watershed. Despite historical records and our efforts to sample all documented locations for leatherside chub, we (and other sampling efforts) were unable to detect extant populations in areas of intermediate distance (Supplemental Fig. 1; Blakney 2012; Meyer et al. 2013). This, in conjunction with the lack of gene flow, known anthropogenic barriers, and observations of a patchy and restricted distribution in many streams, indicates that most populations are completely isolated (Supplemental Fig. 1; Keeley et al. 2012). While population subdivision can occur as a result of migration and homing to specific spawning areas in fishes, leatherside chub are probably much more limited in their migration and homing abilities because of small maximum body size, as observed in other small-bodied fishes (Johnston 2000; Junker et al. 2012). As a result, dispersal between areas may be limited to shorter distances within generations and longer distance gene flow occurring over a multi-generational timeframe, producing a significant isolation by distance relationship that we observed. If population subdivision had occurred prior to and was not a result of recent anthropogenic habitat fragmentation, the significant pattern of isolation by distance would presumably have been lost by the effect of random genetic drift over time.

A recent decline in gene flow is perhaps most evident in the results of our comparison of historical and contemporary rates of migration. Based on estimates of historical

migration, movement rates were estimated to have been about 80 % higher than those for contemporary times. Past studies that have compared migration rates in animal populations embedded in fragmented landscapes have detected differences between historical and contemporary rates of migration for aquatic and semi-aquatic species that have low vagility (Sterling et al. 2012; Apodaca et al. 2012). In other cases, populations show little difference between historical and contemporary rates of migration in terrestrial habitats, indicating that populations had little gene flow even before widespread habitat fragmentation (Chiucchi and Gibbs 2010). Historical rates of migration in leatherside chub indicate that populations were at one time much more inter-connected. Low levels of contemporary gene flow, limited dispersal capability, and the presence of numerous anthropogenic barriers to movement, both between populations and among subbasins, indicate that habitat fragmentation has likely played a role in isolating the remaining populations of this small fish. It may be that in aquatic habitats, unobstructed dispersal corridors are particularly important in maintaining natural levels of genetic diversity and gene flow between populations of small-bodied species with limited abilities to move around, over, or through patches of unsuitable habitat.

Conclusions

Population structure in leatherside chub appears to be representative of both historical vicariance and recent habitat fragmentation. Geologic and mtDNA evidence establish a connection between populations in the three occupied watersheds within the last 600,000 years. Strong patterns of isolation by distance from nuclear DNA and the documentation of the species in main-stem rivers and other currently unoccupied subbasins, argue that the species was more widely distributed, even within the past century. Intense alteration of the landscape has the ability to unravel genetic patterns of structuring that have arisen over many thousands of generations (Neville et al. 2006; Fumagalli et al. 2008). Fragmentation, both natural and anthropogenic, of once-connected populations makes such demes more vulnerable to stochastic extirpation, and because dispersal for aquatic species is severely limited when compared with terrestrial ones, isolation of occupied or suitable habitats makes gene flow and re-colonization impossible (Hanski and Gilpin 1991). Thus the few populations that remain are of high conservation value and efforts should be made to sustain habitat quantity and quality in these streams. The results of our study are in agreement with the growing body of evidence suggesting that human alteration of aquatic systems within the past few hundred years has had a significant negative influence on the distribution, genetic structure and ultimately the survival and maintenance of future adaptability in many fish species (Meffe and Vrijenhoek 1988; McCraney et al. 2010).

Acknowledgments Collection of samples was aided by advice and support of Biologists from the Idaho Department of Fish and Game, Utah Division of Wildlife Resources, Wyoming Department of Game and Fish, Nevada Department of Wildlife, the Bureau of Land Management, and the U.S. Forest Service. We thank J. Luginbill, L. Schultz, P. Thompson, Z. Njoroge, N. Tillotson, R. Bupp, S. Campbell, H. Harris, R. Blackadar and H. Allen for help in the field. D. McPhail, D. Megargle, S. Grunder, K. McAbee, and the Inter-agency leatherside chub Working Group provided helpful advice about sampling and distribution records. Museum specimens and historical records of leatherside chub were made available to us from collections at The University of Michigan (Museum of Zoology), California Academy of Sciences (Institute for Biodiversity Science and Sustainability), College of Idaho (Orma J. Smith Museum of Natural History), Idaho State University (Idaho Museum of Natural History), and the Smithsonian Institution (National Museum of Natural History). G. Smith, D. Nelson, J. Williams, D. Zaroban, D. Catania, and J. Bala helped facilitate the loan of the specimens. We thank S. Lawyer and two anonymous reviewers for comments on an earlier version of this manuscript. Funding for the project was provided by the U.S. Fish and Wildlife Service, facilitated by J. Brostrom, and by the Molecular Research and Core Facility at Idaho State University.

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